Single-Stage Amplifiers

Amplification is an essential function in most analog circuits. We amplify an analog or digital signal because it may be too small to drive a load, overcome the noise of a subsequent stage, or provide logical levels to a digital circuit. Amplification also plays a critical role in feedback systems.

In this chapter, we study the low-frequency behavior of single-stage CMOS amplifiers.

Basic Concepts:

The input-output characteristic of an amplifier is generally a nonlinear function (Fig. 1) that can

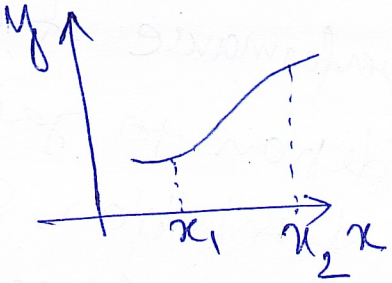


Fig. 1 Input-output characteristic of a nonlinear system.

be approximated by a polynomial over some signal range:

$$y(t) \approx \alpha_0 + \alpha_1 x(t) + \alpha_2 x^2(t) + \cdots + \alpha_n x^n(t) \qquad x_1 \le x \le x_2 \quad \cdots (1)$$

The input and output may be current or voltage quantities. For a sufficiently large range of x,

$$y(t) \approx \alpha_0 + \alpha_1 x(t) \quad \cdots (2)$$

where α_0 can be considered the operating (bias) point and α_1 the small-signal gain. So long as $\alpha_1 x(t) \ll \alpha_0$, the bias point is ~~distri~~ disturbed negligibly, eqn (2) provides a reasonable approximation, and higher order terms are insignificant.

In other words, $\Delta y = \alpha_1 \Delta x$, indicating a linear relationship betn the increments at the input and output. As $x(t)$ increases in magnitude, higher order terms manifest themselves, leading to nonlinearity and necessitating large-signal analysis.

From another point of view, if the slope of the characteristic (the incremental gain) varies with the signal level, then the system is nonlinear.

What aspects of the performance of an amplifier are important? In addition to gain and speed, such parameters as power dissipation, supply voltage, linearity, noise, or maximum voltage swings may be important. Furthermore, the input and output impedances determine how the circuit interacts with preceding and subsequent stages. In practice, most of these parameters trade with each other, making the design a multi-dimensional optimization problem.

Analog design octagon, such trade-offs present many challenges in the design of high-performance amplifiers, requiring intuition and experience to arrive at an acceptable compromise.

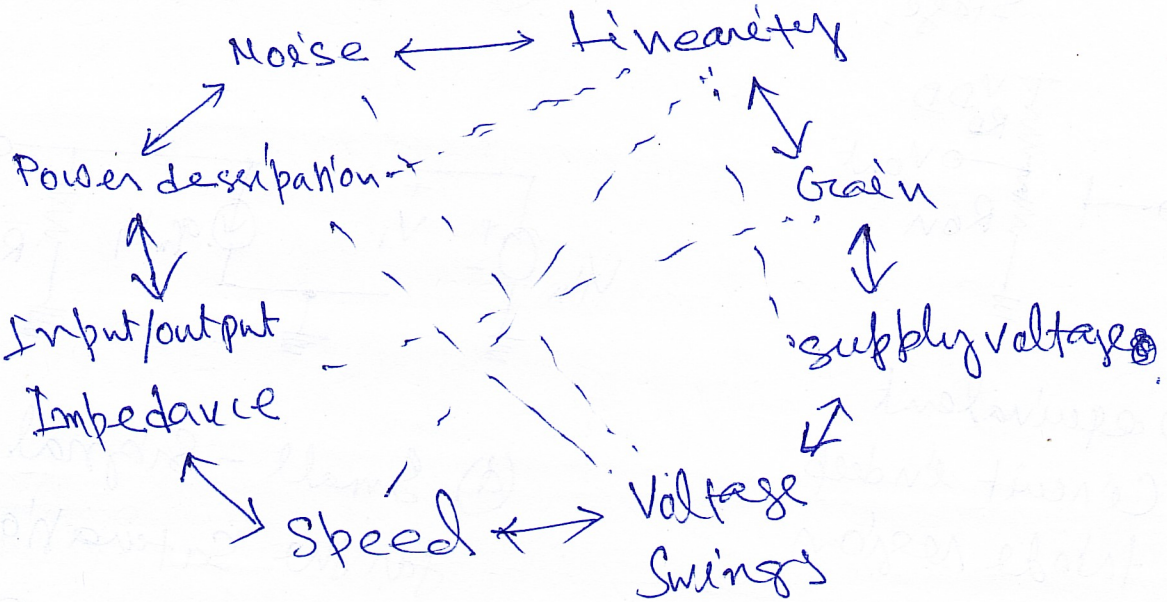
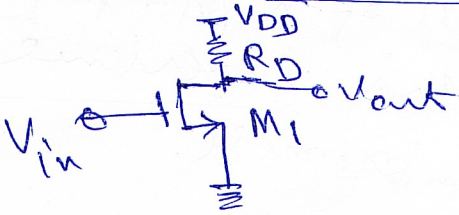
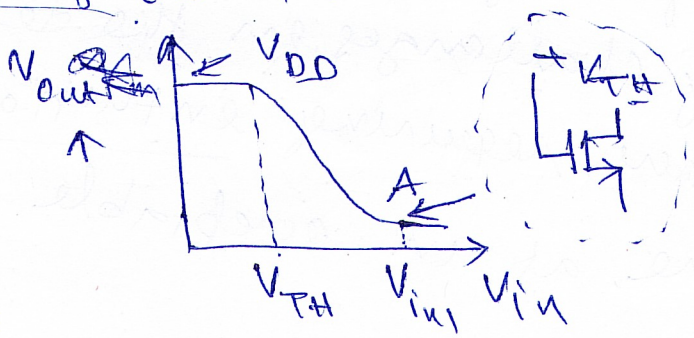


Fig. Analog design octagon.

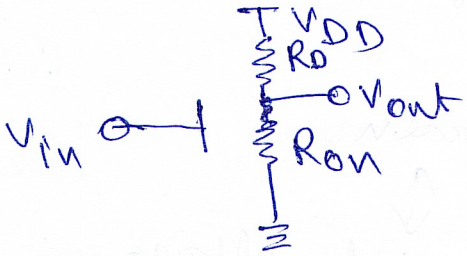
Common-Source Stage with Resistive Load:



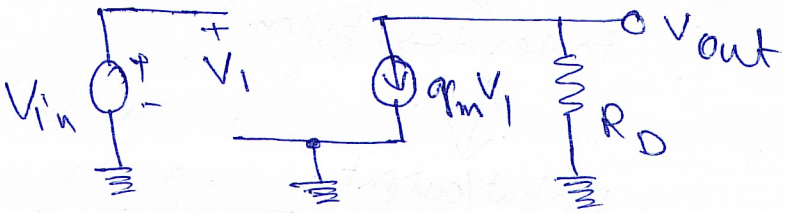
(a) Common-Source Stage.



(b) input-output characteristic



(c) equivalent circuit in deep triode region



(d) Small-Signal model for the saturation region

Fig. 1

By virtue of its transconductance, a MOSFET converts variations in its gate-source voltage to a small-signal drain current, which can pass through a resistor to generate an output voltage.

Shown in Fig. 1(a), the common-source (CS) stage performs such an operation. We study both the large-signal and the small-signal behavior of the circuit. Note that the input impedance of the circuit is very high at low frequencies.

If the input voltage increases from zero, M_1 is off and $V_{out} = V_{DD}$ (Fig. 1.b). As V_{in} approaches V_{TH}, M_1 begins to turn on, drawing current from R_D and

lowering V_{out}. If V_{DD} is not excessively low, M_1 turns on in saturation, and we have

$$V_{out} = V_{DD} - R_D \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{in} - V_{TH})^2 \quad \text{---} (3)$$

where channel-length modulation is neglected. With further increase in V_{in}, V_{out} drops more and the transistor continues to operate in saturation until V_{in} exceeds V_{out} by V_{TH}, [Point A in Fig. 1.b]. At this point,

$$V_{in1} - V_{TH} = V_{DD} - R_D \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{in1} - V_{TH})^2 \quad \text{---} (4)$$

from which $V_{in1} - V_{TH}$ and hence V_{out} can be calculated.

For $V_{in} > V_{in1}$, M_1 is in the triode region:

$$V_{out} = V_{DD} - R_D \frac{1}{2} \mu_n C_{ox} \frac{W}{L} [2(V_{in} - V_{TH}) V_{out} - V_{out}^2] \quad \text{---} (5)$$

If V_{in} is high to drive M_1 into deep triode region, $V_{out} << 2(V_{in} - V_{TH})$, and from the equivalent circuit of Fig. c.

$$V_{out} = V_{DD} \frac{R_{on}}{R_{on} + R_D} \quad \text{---} (6)$$

$$= \frac{V_{DD}}{1 + \mu_n C_{ox} \frac{W}{L} R_D (V_{in} - V_{TH})} \quad \text{---} (7)$$

Since the transconductance drops in the triode region, we usually ensure that $V_{out} > V_{in} - V_{TH}$, operating to the left of point A in Fig. 1.6. Using eqn (5) as the input-output characteristic and viewing its slope as the small-signal gain, we have

$$A_v = \frac{\partial V_{out}}{\partial V_{in}} \quad \text{.....} (8)$$

$$= -R_D \mu_n C_{ox} \frac{W}{L} (V_{in} - V_{TH}) \quad \text{...} (9)$$

$$= -g_m R_D \quad \text{.....} (10)$$

This result can be directly derived from the observation that M_1 converts an input voltage change ΔV_{in} to a drain current change $g_m \Delta V_{in}$, and hence an output voltage change $-g_m R_D \Delta V_{in}$. The small-signal model Fig. 1d yields the same result.

Even though derived for small-signal operation, the eqn $A_v = -g_m R_D$ predicts certain effects if the circuit senses a large signal swing. Since g_m itself varies with the input signal according to $g_m = \mu_n C_{ox} (W/L)(V_{in} - V_{TH})$, the gain of the circuit changes substantially if the signal is large. In other words, if the gain of the circuit varies significantly with the signal swing, then the circuit operates in the large-signal model. The dependence of the gain upon the signal level leads to nonlinearity, usually an undesirable effect.

A key result here is that to minimize the nonlinear the gain equation must be a weak function of signal-dependent parameters such as g_m.

How do we maximize the voltage gain of a common-source stage? writing as

$$A_V = -\sqrt{2\mu_n C_{ox}\frac{W}{L} I_D}\,\frac{V_{RD}}{I_D} \quad ----(11)$$

Where V_{RD} denotes the voltage drop across R_D, we have

$$A_V = -\sqrt{2\mu_n C_{ox}\frac{W}{L}}\,\frac{V_{RD}}{\sqrt{I_D}} \quad ----(12)$$

Thus, the magnitude of A_v can be increased by increasing $\frac{W}{L}$ or V_{RD} or decreasing I_D if other parameters are constant. It is important to understand the trade-offs resulting from this equn. A larger device size leads to greater device capacitances, and a higher V_{RD} limits the maximum voltage swings. For example, if $V_{DD} - V_{RD} = V_{in} - V_{TH}$, Then M_1 is at the edge of the triode region, allowing only very small swings at the (output and input). If V_{RD} remains constant and I_D is reduced, then R_D must increase, thereby leading to a greater time constant at the output node. In other words, as noted in the analog design octagon, the circuit exhibits trade-offs bet^n gain

bandwidth, and voltage swings. Lower supply voltages further tighten these trade-offs.

For large values of R_D, the effect of channel length modulation in M_1 becomes significant. Modifying eqn (9) to include this effect,

$$V_{out} = V_{DD} - R_D \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{in} - V_{TH})^2 (1 + \lambda V_{out}), \quad \text{---(13)}$$

we have

$$\frac{\partial V_{out}}{\partial V_{in}} = -R_D \mu_n C_{ox} \frac{W}{L} (V_{in} - V_{TH})(1 + \lambda V_{out}) - R_D \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{in} - V_{TH})^2 \lambda \frac{\partial V_{out}}{\partial V_{in}} \quad \text{---(14)}$$

Using approximation $I_D \approx (1/2) \mu_n C_{ox} (W/L) (V_{in} - V_{TH})^2$, we obtain

$$A_V = -R_D g_m - R_D I_D \lambda A_V \quad \text{---(15)}$$

and hence

$$A_V = -\frac{g_m R_D}{1 + R_D \lambda I_D} \quad \text{---(16)}$$

Since $\lambda I_D = 1/r_o$,

$$A_V = -g_m \frac{r_o R_D}{r_o + R_D} \quad \text{---(17)}$$

The small-signal model of below gives the same result with much less effort. That is, since $g_m V_1 (r_o \| R_D) = -V_{out}$ and $V_1 = V_{in}$, we have $V_{out}/V_{in} = -g_m(r_o \| R_D)$. Note that, as mention V_{in}, V_1 and V_{out} in this figure denote small-signal quantities.

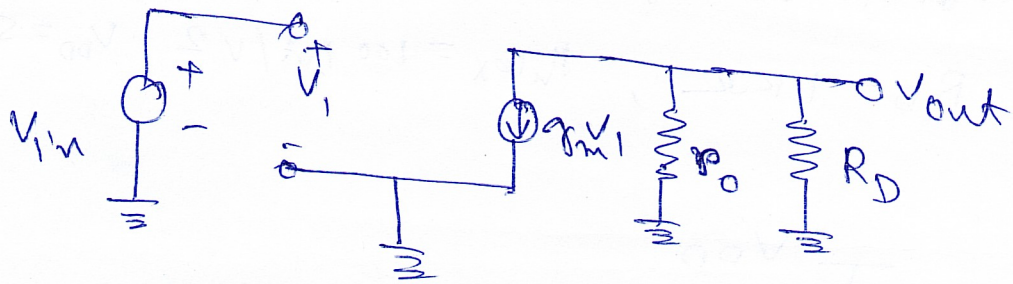
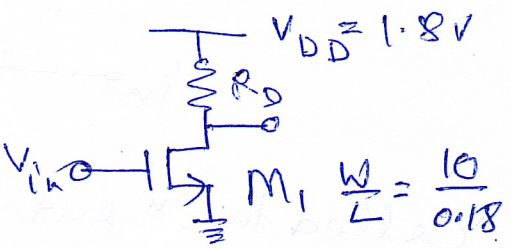


Fig. Small-signal model of CS stage including the transistor output resistance.

Problem: ① Calculate the small-signal voltage gain of the CS stage shown in Fig below if $I_D = 1\,mA$, $\mu_n C_{ox} = 100\,\mu A/V^2$, $V_{TH} = 0.5V$, and $\lambda = 1.0$. Verify that M_1 operates in saturation. $R_D = 1K\Omega$



Sol^n: We have $g_m = \sqrt{2\mu_n C_{ox}\frac{W}{L} I_D} = \frac{1}{300\Omega}$

Thus $A_V = -g_m R_D = -3.33$.

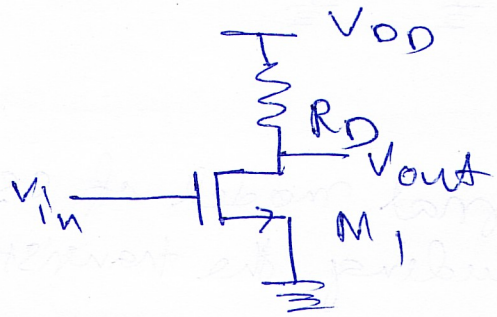
To check the operation region, we first determine the gate-source voltage: $V_{GS} = V_{TH} + \sqrt{\frac{2I_D}{\mu_n C_{ox}\frac{W}{L}}}$

$= 1.1V$

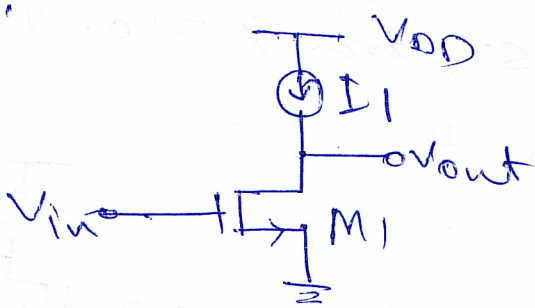
The drain voltage is equal to $V_{DD} - R_D I_D = 0.8\,V$.
Since $V_{GS} - V_{TH} = 0.6\,V$, the device indeed operates in saturation and has a margin of $0.2\,V$ with respect to the triode region. For example, if R_D is doubled with the intention of doubling A_V, then M_1 enters the triode region and its transconductance drops.

(2) For MOSFET circuit shown in Fig. 2 determine the small-signal voltage gain if $\left(\frac{W}{L}\right) = 50/0.5$

$I_D = 0.5\,mA$ $R_D = 1\,k\Omega$, $\mu_n C_{ox} = 100\,\mu A/V^2$, $V_{DD} = 5\,V$
$V_{TH} = 0.5\,V$



(3) Assuming M_1 in Fig. below is biased in saturation, calculate the small-signal gain of the circuit.



Sol'n: Since I_1 ~~increases~~ introduces an infinite impedance, the gain is limited by the output resistance of M_1: $A_V = -g_m r_o$

Called the "intrinsic gain" of a transistor, this quantity represents the maximum voltage gain that can be achieved using a single device. In today's CMOS technology, $g_m r_o$ of short-channel devices is between roughly 10 and 30. Thus, we usually assume $1/g_m \ll r_o$.

In Fig. Kirchhoff's current law (KCL) requires that $I_{D_1} = I_1$. Then, how can V_{in} change the current of M_1 if I_1 is constant? Writing the total drain current of M_1 as

$$I_{D_1} = \frac{1}{2}\mu_n C_{ox}\left(V_{in} - V_{TH}\right)^2\left(1 + \lambda V_{out}\right)$$
$$= I_1$$

We note that V_{in} appears in the square term and V_{out} in the linear term. As V_{in} increases, V_{out} must decrease such that the product remains constant. We may nevertheless say "I_{D_1} increases as V_{in} increases". This statement simply refers to the quadratic part of the equation.